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Tracking control of inverted pendulum-inspired robotic systems using optimized integrator-based LQR

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Abstract
The principles of inverted pendulum systems are extensively utilized in applications such as slosh modeling, humanoid robots, and space technologies. Consequently, design of effective controllers for these systems is a critical task. This study focuses on the inverted pendulum models through linearization and design of two controllers which are the linear quadratic regulator with particle swarm optimization and the linear quadratic regulator with integrator based on particle swarm optimization. The optimal controller gains are derived using particle swarm optimization, a widely used artificial intelligence technique. The proposed controllers are tested against various reference inputs including impulse, step, square, sinusoidal, and sawtooth signals on four different inverted pendulum models. The controllers are also applied to Cubli rovers and monowheel robots which utilize inverted pendulum principles. Comparative analysis is conducted with existing controllers from literature including the neuro-fuzzy controller, genetic algorithm-based linear quadratic regulator, and whale optimization-based sliding mode controller. Further evaluation involves testing the controllers on an experimental inverted pendulum setup, both on the nominal system and with an uncounted external weight or bob mounted to the nominal system, introducing unmodeled dynamics. The robustness of the controllers is assessed in this augmented system. The performance of these proposed controllers are quantified using error indices and the absolute maximum peak. Results and comparative study indicate that the proposed controllers effectively regulate system dynamics, minimize errors, and maintain system stability. These findings highlight the controllers' versatility and efficacy in various inverted pendulum-inspired applications.

Keywords Inverted pendulum · Linear quadratic regulator · Particle swarm optimization · Cubli rover · Monowheel

Abbreviations

V_m	Nominal input voltage, 6 V
K_g	Planetary gearbox gear ratio, 3.71
M	Mass of cart, 0.75 Kg
M_p	Pendulum mass, 0.127 Kg
J_m	Rotor moment of inertia, 3.90×10^{-7}kg m^2
l_p	Medium pendulum length, 0.1778 m
r_{mp}	Motor pinion radius, 6.35×10^{-3} m
β_{eq}	Equivalent viscous coefficient, 5.4 N.m.s/rad
β_p	Viscous damping coefficient, 0.0024 N.m.s/rad
I_p	Medium pendulum moment, 1.20×10^{-3} kg m^2
g	Gravitational constant, 9.81 m/s^2

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1 Introduction

Inverted pendulum is a widely popular control problem involving balancing of a pendulum in the upright position contrary to the traditional pendulum which remains in the downward direction [1]. In the recent years, the inverted pendulum concept was applied in balancing robots, segway vehicles [2], vertical propulsive rocket landing [3]. Controlling an inverted pendulum poses several challenges due to its inherently unstable nature. The major challenges are nonlinearity, high sensitivity to external factors, limited actuation [1]. To

address these challenges, several controllers were proposed in the past decades. Some widely used controllers are linear quadratic regulator (LQR), proportional–integral–derivative (PID) controller, sliding mode controller [4].

In recent literature, several works were carried out in the inverted pendulum setup. A sliding mode control for a single inverted pendulum (SIP) was proposed in [5]. An adaptive dynamic programming-based optimal control for SIP was presented in [6]. In [7], the authors proposed a holistically optimized controller for inverted pendulum setup. An Newton–Raphson method-based LQR controller for inverted pendulum was presented in [8]. A PID controller for a single inverted pendulum based on the state-space model was presented in [9]. In [10], a decision-making approach for the control of inverted pendulum was presented with a neuro-fuzzy controller (NFC), a indirect adaptive controller (IAC), and a PID controller. The authors in [11] presented a variable structure control (VSC) along with a PID controller for controlling an inverted pendulum. The authors in [12] proposed a robust LQR controller for inverted pendulum. A Lagrangian equation-based advanced and reliable LQR controller was presented in [13], to reduce the variation in cart angle and pendulum angle for single inverted pendulum and to obtain optimal controller gain of the LQR controller metaheuristic approaches are used [14]. To control a single inverted pendulum, a reduction in penalty functions for NM simplex-based LQR tuning was proposed in [15]. A fuzzy logic-based control for a single inverted pendulum was proposed in [16]. Recent research explores a range of nonlinear and intelligent control strategies aimed at enhancing robustness and reducing swing in underactuated pendulum systems [17, 18]. Approaches such as fuzzy LQR, neural–fuzzy control, and reinforcement learning-based methods have shown promising results in improving stability and control performance [19–22]. While our current work emphasizes PSO-optimized linear controllers, these nonlinear and learning-based techniques offer valuable benchmarks and form a strong foundation for future extensions. A brief outline of controllers and optimization techniques used in inverted pendulum applications are presented in Table 1.

1.1 Applications of the principles of the inverted pendulum

The inverted pendulum is a well-known problem in dynamics and control theory, commonly used to demonstrate feedback control and stability principles in diverse systems, including slosh modeling, thrust vectoring during vertical launch, VTOL aircraft, robotic arms, and humanoid robots [35]. Slosh modeling examines the dynamics of moving liquids within a container, which is essential for applications like spacecraft fuel tanks, liquid transport vehicles, and biomechanics [36, 37]. An outline of some applications of inverted pendulum is presented in Fig. 1. In the context of spacecraft like the Tracking and Data Relay Satellite (TDRS), researchers have examined the dynamics of liquid sloshing in both upright and inverted bladdered tanks. The inverted tank, where the liquid is positioned above the bladder, shows unconventional sloshing due to its highly asymmetrical orientation. To accurately model this behavior, a mechanical model resembling an inverted pendulum, which can handle small oscillations and large reorientations, is necessary [38–40].

Inverted pendulum models have been utilized in biomechanics to study the craniomandibular complex and cervical spine [41], analyze lateral dynamic weight shifts [42], and simulate the dynamics of quiet human stance [43]. These models aid in comprehending the balance and control strategies used by the human body in different postures and movements.

Robotic systems frequently use inverted pendulum models for motion control. For instance, neural network-based motion control has been applied to an underactuated wheeled inverted pendulum model [44], and push recovery strategies have been implemented in humanoid robots using a linear inverted pendulum model [45]. The Cubli rover is an intriguing robotic invention that employs the principles of the inverted pendulum to accomplish impressive feats such as jumping, balancing, and autonomous movement [46]. One of the most exciting uses of Cubli technology is in space exploration. Its ability to balance, jump, and move autonomously makes it an excellent candidate for exploring the surfaces of planets and asteroids [46]. The principles of the inverted pendulum are utilized by monowheel robots, which balance on a single wheel, to maintain stability and maneuverability [47]. Autonomous transportation in various settings can be achieved using monowheel robots. Their ability to balance and navigate through tight spaces makes them ideal for use in shopping malls, offices, airports, and other crowded environments [47]. These inverted pendulum models are essential for creating robots capable of maintaining balance and recovering from perturbations.

1.2 Objectives of this work

In this work, a particle swarm optimization (PSO)-based linear quadratic regulator (LQR) and its variant with an integrator (LQRI) are proposed for inverted . The key objectives of this work are given as follows:

- The linearized system dynamics for the nonlinear single inverted pendulum are presented.
- The system with LQR and LQRI controllers is formulated.

Table 1 A brief outline of the existing literature based on the controllers and optimization techniques used in relation to the control of the inverted pendulum systems

Category	Process	References															
		[8]	[9]	[10]	[11]	[23]	[24]	[25]	[26]	[27]	[28]	[29]	[30]	[31]	[32]	[33]	[34]
Controller	Linear quadratic regulator	✓											✓				
	Linear quadratic Gaussian										✓						
	Proportional–integral–derivative	✓	✓	✓			✓	✓	✓	✓						✓	
	Sliding mode				✓	✓				✓							
	Nonlinear auto-regressive							✓									
	Fuzzy-based			✓									✓	✓		✓	✓
	Model reference control														✓		
Optimization techniques	Quadratic programming													✓			
	Gradient-based								✓								
	Genetic algorithm	✓										✓					
	Particle swarm optimization															✓	✓
	Binary search algorithm										✓						
	Adaptive learning algorithm			✓				✓									
	Whale-based algorithm									✓							
	Neural network-based algorithm			✓		✓									✓		
	Artificial bee colony algorithm						✓										

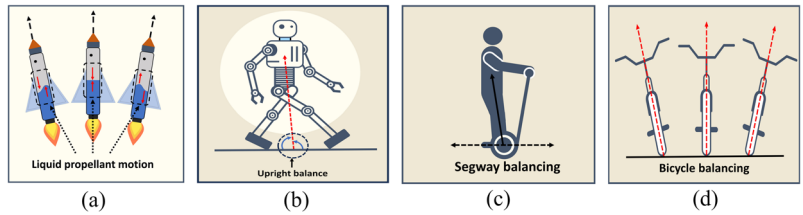


Fig. 1 Applications of the principles of inverted pendulum: **a** slosh modeling in launch vehicles; **b** balancing of walking robots; **c** balancing in segways; **d** bicycle movement and its balance

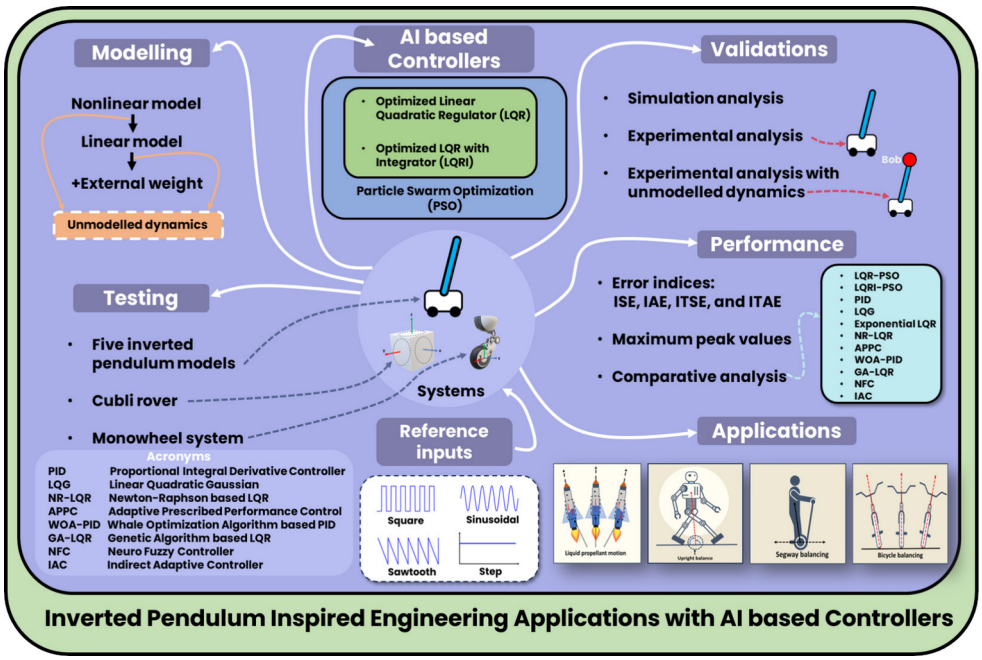


Fig. 2 An outline of this work

- The optimized weighting matrices and the controller gains are obtained using particle swarm optimization (PSO) for both the LQR and LQRI controller.
- The proposed LQR-PSO and LQRI-PSO controllers are tested in the simulations for the single inverted pendulum for impulse-, step-, square-, sinusoid-, and sawtooth-type reference inputs.
- The effectiveness of the proposed controllers is tested on four different inverted pendulum models, Cubli rover, and monowheel robot applications where inverted pendulum principles are utilized.
- The proposed LQR-PSO and LQRI-PSO controllers are experimentally validated on the hardware setup of an inverted pendulum.
- The inverted pendulum setup is augmented with an external weight (bob) which adds unmodeled dynamics to the system and the proposed controllers are tested on the modified system to validate robustness.
- For the modified system, the experimental validations are conducted for square-, sinusoid-, and sawtooth-type reference inputs.
- For all the experiments conducted, the error indices like ISE, IAE, ITSE, ITAE, and the absolute maximum peak $|M_p|$ values for input voltage (V_m), cart position (x_c), and pendulum angle (α) are presented.
- An extensive comparative study is presented to validate the proposed work.

An outline of this work is presented in Fig. 2. The paper is organized as follows: Section 2 presents the single-inverted-pendulum modeling followed by the LQR-PSO and the LQRI-PSO controller design in Sect. 3, and Sect. 4 presents the simulation and the experimental results and observations, which is followed by the conclusion and future aspects in Sect. 5.

2 Single inverted pendulum

The system in Fig. 3 represents a simple linear inverted pendulum mounted on a moving cart. x_c is the cart position, and α is represented as the pendulum angle in the direction of counterclockwise. l_p denotes the length of the pendulum from pivot to center of gravity. m_p and J_p denote the mass of the pendulum and the moment of inertia about the center of mass.

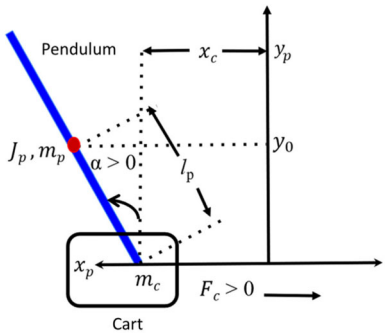


Fig. 3 Schematic diagram of a single inverted pendulum on a moving cart

2.1 Equations of motion

The system in Fig. 1 is a single-inverted-pendulum system having two DOF in the form of x_c, α, and one actuation as the input voltage to the motor connected to the cart (V_m). The coordinates (x_p, y_p) indicate the location of the pendulum's center of gravity which are defined as

$$x_p = x_c - l_p \sin(\alpha) \tag{1}$$
$$y_p = l_p \cos(\alpha) \tag{2}$$

By differentiating the (1) and (2), we will get

$$\dot{x}_p = \dot{x}_c - l_p \cos(\alpha)\dot{\alpha} \tag{3}$$
$$\dot{y}_p = -l_p \sin(\alpha)\dot{\alpha} \tag{4}$$

There is never any vertical displacement in the cart's linear motion as it is horizontal. Therefore, the gravitational potential energy of the pendulum, as described below, completely represents the whole potential energy:

$$V_T = M_p g l_p \cos(\alpha) \tag{5}$$

The total kinetic energy of the system is the sum of the cart and pendulum's kinetic energy. Cart and pendulum both have translational and rotational kinetic energy due to their movement. The translational kinetic energy of cart is T_{ct}:

$$T_{ct} = \frac{1}{2} M \dot{x}_c^2 \tag{6}$$

The cart's rotational kinetic energy due to the cart's DC motor is T_{cr} which is defined as follows:

$$T_{cr} = \frac{1}{2} \frac{J_m K_g^2 \dot{x}_c^2}{r_{\mathrm{mp}}^2} \tag{7}$$

So, total cart's kinetic energy (T_c) is $T_{ct} + T_{cr}$:

$$T_c = \frac{1}{2} M_c \dot{x}_c^2 \tag{8}$$

where $M_c = M + \frac{J_m K_g^2}{r_{\mathrm{mp}}^2}$.
The pendulum translational kinetic energy can be written as:

$$T_{pt} = \frac{1}{2} M_p (\dot{x}_p^2 + \dot{y}_p^2) \tag{9}$$

The rotational energy of a pendulum is defined as:

$$T_{pr} = \frac{1}{2} I_p \dot{\alpha}^2 \tag{10}$$

The total kinetic energy of pendulum can obtain using (9) and (10):

$$T_p = \frac{1}{2}[M_p(\dot{x}_p^2 + \dot{y}_p^2) + I_p \dot{\alpha}^2] \tag{11}$$

Therefore, we can compute the total kinetic energy by (8) and (11)

$$T_T = \frac{1}{2}(M_c + M_p)\dot{x}_c^2 - M_p l_p \cos(\alpha)\dot{\alpha}\dot{x}_c + \frac{1}{2}(I_p + M_p I_p^2)\dot{\alpha}^2 \tag{12}$$

The Lagrangian of the system is given by

$$L = T_T - V_T \tag{13}$$

Applying the Euler–Lagrangian energy equation to the system

$$\frac{\partial^2 L}{\partial t \partial \dot{x}_c} - \frac{\partial L}{\partial x_c} = Q_{x_c} \tag{14}$$
$$\frac{\partial^2 L}{\partial t \partial \dot{\alpha}} - \frac{\partial L}{\partial \alpha} = Q_\alpha \tag{15}$$

where Q_{x_c} and Q_α are the generalized force applied on the x_c and x_α, respectively,

$$Q_{x_c} = F_c - \beta_{\mathrm{eq}} \dot{x}_c \tag{16}$$
$$Q_\alpha = -\beta_p \dot{\alpha} \tag{17}$$

From (11), (12), and (13), Lagrangian of system can be written as,

$$L = \frac{1}{2}(M_c + M_p)\dot{x}_c^2 - M_p l_p \cos(\alpha)\dot{\alpha}\dot{x}_c + \frac{1}{2}(I_p + M_p I_p^2)\dot{\alpha}^2 - M_p g l_p \cos(\alpha) \tag{18}$$

Nonlinear equation of motions can be obtained by (14) and (15)

$$F_c - \beta_{\text{eq}}\dot{x}_c = (M_c + M_p)\ddot{x}_c - M_p l_p \cos(\alpha)\ddot{\alpha} + M_p l_p \sin(\alpha)\dot{\alpha}^2 \quad (19)$$

$$-\beta_p\dot{\alpha} = -M_p l_p \cos(\alpha)\ddot{x}_c + M_p g l_p \sin\alpha \quad (20)$$

2.2 Linearized model

Around the equilibrium point (upright pendulum), the nonlinear model is linearized so that $\sin(\alpha) \cong \alpha$ and $\cos(\alpha) \cong 1$ [12]. The system's mathematical model in state-space form is necessary for the construction of the state feedback controller. The system's state-space equations are obtained by linearizing the equations of motion [48]. The state-space equations are

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (21)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} \quad (22)$$

The system matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, the state vector $\mathbf{x} \in \mathbb{R}^n$, the input matrix $\mathbf{B} \in \mathbb{R}^{n\times p}$, the input $u \in \mathbb{R}^p$, the output vector $\mathbf{y} \in \mathbb{R}^q$, and the output matrix $\mathbf{C} \in \mathbb{R}^{n\times q}$. For utilized system, the order (n) is 4, the number of input (p) is 1, and the number of output (q) is 2. The state-space vector of the given system is

$$\mathbf{x} = [x_c, \alpha, \dot{x}_c, \dot{\alpha}]^T \quad (23)$$

3 Controller design

The LQR approach is a potent way to create controllers for intricate systems with demanding performance standards since it looks for the best controller to minimize a specified cost function.

3.1 System analysis with LQR controller

For the given system, the cart must maintain a specific cart position while also balancing the single inverted pendulum. The intended cart position is represented by x_d, which is taken as the desired cart position. Our main objective is to reduce the error between the intended cart position (x_d) and the present cart position (x_c). To minimize this error, the optimal controller gains are obtained using the PSO technique. The following system was utilized for the proposed analysis [48]. The system matrix (**A**), the input matrix (**B**), and the output matrix (**C**) are

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.8703 & -11.9241 & -0.0094 \\ 0 & 46.2580 & -51.6449 & -0.5012 \end{bmatrix} \quad (24)$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1.5662 \\ 6.7834 \end{bmatrix} \quad (25)$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (26)$$

The controller output is

$$u = -\mathbf{K}(\mathbf{x_r} - \mathbf{x}) \quad (27)$$

where $\mathbf{x_r}$ is the reference input which is defined as

$$\mathbf{x_r} = \begin{bmatrix} x_d & 0 & 0 & 0 \end{bmatrix}^T \quad (28)$$

The controller is designed to guide the cart to follow the desired position of the cart and to stabilize the pendulum in an upright position. The proposed controller aims to mitigate these effects by compensating for unmodeled system dynamics. The LQR controller gives a gains **K** by using a quadratic cost function to reduce the error and mitigate a unmodeled system dynamics. The cost function for LQR controller is defined as

$$J = \int_0^\infty (\mathbf{x}^T\mathbf{Q}\mathbf{x} + u^T R u)dt \quad (29)$$

where **Q** and R are the positive semi-definite matrices known as the weighting matrices for the LQR cost function. These matrices will do a trade-off on the system performance and the system input as **Q** penalizes the state vector and R penalizes the system output. The controller gain **K** can obtained as

$$\mathbf{K} = R^{-1}\mathbf{B}P \quad (30)$$

The parameter $P>0$ is the solution of the algebraic Riccatti equation given as

$$\mathbf{A}^T P + P\mathbf{A} - P\mathbf{B}R^{-1}\mathbf{B}^T P + \mathbf{Q} = 0 \quad (31)$$

3.2 System analysis with LQRI controller

To further improve the performance of the LQR controller, an integral-based LQR controller (LQRI) is presented. The integral between the intended cart position (x_d) and the present

cart position (x_c) is augmented to the modified state vector [48, 49]. The system is modified as

$$\dot{\hat{\mathbf{x}}} = \begin{bmatrix} \mathbf{A} & 0 \\ 1 & 0 \end{bmatrix} \hat{\mathbf{x}} + \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} u$$
$$= \hat{\mathbf{A}}\hat{\mathbf{x}} + \hat{\mathbf{B}}u \tag{32}$$
$$\mathbf{y} = \hat{\mathbf{C}}\hat{\mathbf{x}} \tag{33}$$

The modified state-space vectors are given as

$$\hat{\mathbf{x}} = \left[x_d - x_c, \alpha, \dot{x}_c, \dot{\alpha}, \int (x_d - x_c)\, dt \right]^T \tag{34}$$

The modified system matrix ($\hat{\mathbf{A}}$), input matrix ($\hat{\mathbf{B}}$), and the output matrix ($\hat{\mathbf{C}}$) are given as

$$\hat{\mathbf{A}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0.8703 & -11.9241 & -0.0094 & 0 \\ 0 & 46.2580 & -51.6449 & -0.5012 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{35}$$

$$\hat{\mathbf{B}} = \begin{bmatrix} 0 \\ 0 \\ 1.5662 \\ 6.7834 \\ 0 \end{bmatrix} \tag{36}$$

$$\hat{\mathbf{C}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{37}$$

The controller output is given as

$$u = -\mathbf{K}(\mathbf{x_r} - \hat{\mathbf{x}}) \tag{38}$$

where $\mathbf{x_r}$ is the reference input which is defined as

$$\mathbf{x_r} = \left[x_d \; 0 \; 0 \; 0 \right]^T \tag{39}$$

The cost function J for the LQR controller is given as

$$J = \int_0^\infty (\hat{\mathbf{x}}^T \mathbf{Q} \hat{\mathbf{x}} + u^T R u) dt \tag{40}$$

The weighing matrices **Q**, **R**, and the controller gains **K** are obtained using Eqs. 30 and 31 (Fig. 4).

3.3 Design of PSO-based LQR and LQRI controllers

The particle swarm optimization (PSO), developed by James Kennedy and Russell C Eberhart in 1995, is an intelligent optimization algorithm that is inspired by the social behavior of birds and fishes. In the PSO technique, each particle has its position and velocity. Particles move at a certain position and velocity, much like a bird looking for the best path. A particle swarm structure like a bird swarm is formed by many particles. The PSO method, like other evolutionary algorithms, starts with initializing the particle swarm, searches for local best and global best, updates the position and velocity accordingly, evolves to the next generation, and repeats the process till the best solution is achieved or maximum number of iteration is reached.

PSO has been widely used in various fields including economic load dispatch, power drive, task assignment in distributed systems, and vehicle routing problems. These widespread applications of PSO have motivated this study to adopt the PSO algorithm [50].

The weighting matrices **Q** and **R** are required to calculate the gain **K** of the LQR controller. The agent's positions are represented by the value of **Q** and **R** in PSO. For LQR-PSO controller, **Q** = diag(Q_1, Q_2, Q_3, Q_4) is a (4 × 4) diagonal matrix and R=R_1 is a scalar quantity as the number of input is 1. For the LQRI-PSO controller, where **Q** = diag(Q_1, Q_2, Q_3, Q_4, Q_5) is a (5 × 5) diagonal matrix and R=R_1, the position of the particle i at iteration n can be written as, in PSO

Algorithm 1 The pseudocode of the proposed method.

Ensure: $\mathbf{Q_i} \geq 0 \quad \forall \; i = 1, 2, .., 4, R_1 > 0$
1: $Initialize : x_d, \alpha_d$
2: $Initialize : X_i(n)$ refer to equation (41), for the LQR-PSO and LQRI-PSO controller use equation (42) and (43) respectively.
3: **while** $n \leq n_{max}$ **do**
4: **while** $i \leq N$ **do**
5: Get the controller gains **K** from the system matrix **A**, input matrix **B**,
and the Riccatti equation by using equations (24) (25) (29) (30) (31)
6: Get x_{c_i}, α dynamics from the linearized inverted pendulum model using
equations (21) to (28)
7: Get the cost function $C_i(n)$ from the desired and actual cart position
and pendulum angle using equation (44).
8: Get the local best particle $x_i^{lb}(n)$ and the local best value C_i^{lb} using
equation (46) and (47) respectively
9: Get the global best particle index (r, s) and value $(x^{gb}(n))$ using
equations (48) and (49)
10: $update$: the particles velocity $V_{id}(n+1)$ and positions $(Xi(n+1))$ using
equations (50) and (51) respectively
11: **end while**
12: **end while**
13: Get the global best particle index (r, s) and value $(x^{gb}(n))$ using equations (48) and (49)
14: Get $X_r(s)$, which gives the optimal $\mathbf{Q_i}$ and R
15: Get the optimal controller gain **K** using equations (24) (25) (29) (30) (31)
16: Get the optimal x_{c_i}, α dynamics from the linearized inverted pendulum model using equations (21) to (28)

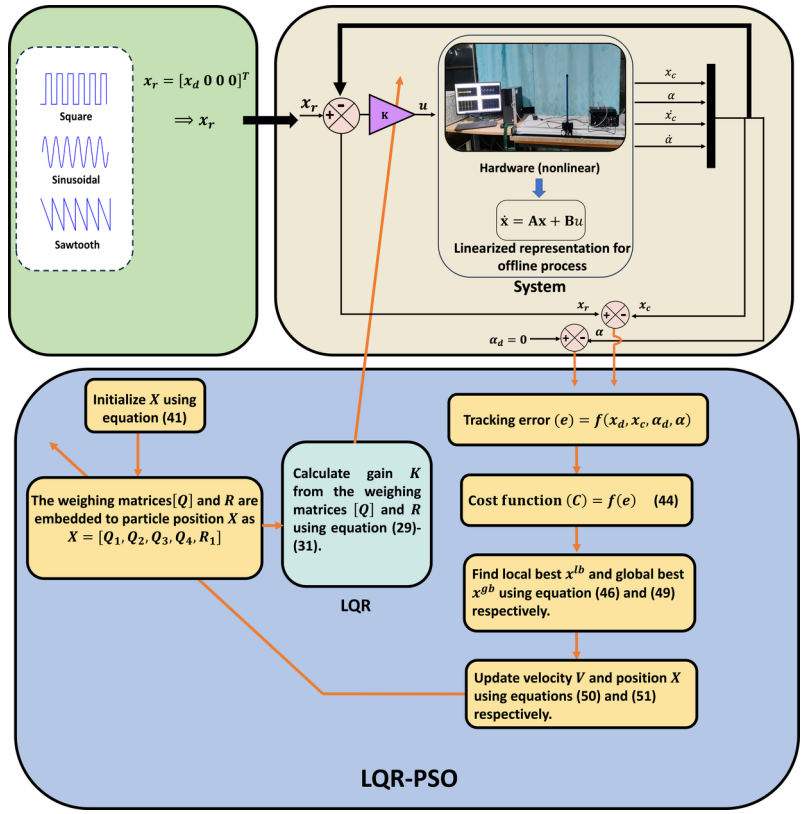


Fig. 4 An outline of the LQR-PSO controller design for different reference inputs

$$X_i(n) = [x_{i1}(n), \ldots, x_{id}(n)] \ \forall \ i = 1, 2, \ldots, N \tag{41}$$

In Eq. (41), N and d represent the total number of particles and the dimension of search space, respectively. For the LQR-PSO controller, $d = 5$ such that $x_{i1} = Q_{i1}, x_{i2} = Q_{i2}, x_{i3} = Q_{i3}, x_{i4} = Q_{i4}$, and $x_{i5} = R_{1i}$. For the LQRI-PSO controller, $d = 6$ such that $x_{i1} = Q_{i1}, x_{i2} = Q_{i2}, x_{i3} = Q_{i3}, x_{i4} = Q_{i4},, x_{i5} = Q_{i5}$, and $x_{i6} = R_{1i}$. Initial velocity for all the particles is assigned to zero. The particle positions are randomly generated considering

$$Q_j \geq 0 \ \forall \ j = 1, \ldots, 4 \text{ and } R_1 > 0 \text{ for LQR-PSO} \tag{42}$$

$$Q_j \geq 0 \ \forall \ j = 1, \ldots, 5 \text{ and } R_1 > 0 \text{ for LQRI-PSO} \tag{43}$$

The objective is to minimize the error for both the cart position and the pendulum angle. To achieve this, we consider the cost function for the PSO algorithm defined as

$$C_i(t) = \int_0^{t_f} ((x_d - x_{c_i})^2 + (\alpha_d - \alpha_i)^2)dt \tag{44}$$

where $C_i(n)$ is represented the cost value for ith particle at nth iteration, α_d is the desired pendulum angle which is zero, and t_f is the total simulation time for the system.
PSO technique used the local best particle and the global best particle to update the position and velocity of particles. The cost can be calculated using Eq. (44) for each particle. The local best particle can be obtained using the following index

$$l = \text{index}\left[\min_{i \in 1, \ldots, N} C_i(n)\right] \tag{45}$$

The local best particle is defined as

$$x_i^{lb}(n) = X_l(n) \tag{46}$$

The local best value $(C_i^{lb}(n))$ can be calculated as

$$C_i^{lb}(n) = C_l(n) \tag{47}$$

An index of global best particle can be calculated as

$$(r, s) = \text{index}\left[\min_{\substack{i \in 1, \ldots, N \\ n \in 1, \ldots, n_{\max}}} C_i^{lb}(n)\right] r \in [1, \ldots, i] \text{ and } s \in [i, \ldots, n] \tag{48}$$

where $n_{\max}$ is the maximum number of iteration
Therefore, the global best particle x^{gb} can be obtained as

$$x^{gb}(n) = X_r(s) \quad r \in i \text{ and } s \in n \tag{49}$$

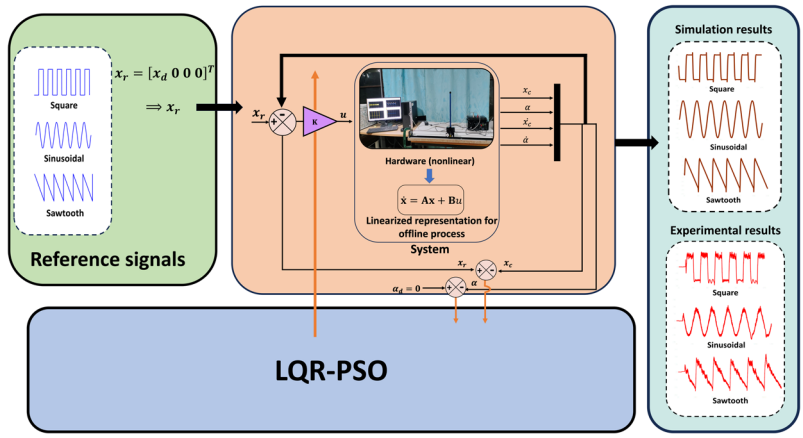


Fig. 5 An outline of the results obtained through LQR-PSO controller

Table 2 Error indices for the tracking simulations for the cart position x_c under a square-type reference input

Error indices	LQR-PSO	LQRI-PSO
	$t = 0s - 120$ s	$t = 0s - 120$ s
ISE	3.093003	2.990400
IAE	6.041120	5.796219
ITSE	182.4427	176.3504
ITAE	349.5883	335.2698

By the iterative process, the PSO method updates the particle velocity and particle position using the following equations

$$V_{id}(n+1) = \omega V_{id}(n) + c_1 \text{rand}_1 (x_{id}^{lb}(n) - x_{id}(n)) + c_2 \text{rand}_2 (x^{gb}(n) - x_{id}(n)) \quad (50)$$

$$X_i(n+1) = X_i(n) + V_{id}(n+1) \quad (51)$$

where $V_{id}(n)$ is velocity of dth dimension of ith particle at nth iteration, $x_{id}(n)$ is position of dth dimension of ith particle at nth iteration, ω is inertia coefficient, c_1 is personal acceleration coefficient, c_2 is social acceleration coefficient, and rand_1 and rand_2 are the random values between 0 and 1. After the updation of particle position and velocity, we will again calculate the cost function using Eq. (44) and update the local best and global best particle accordingly using (45) to (49). After completion of $n_{\max}$ iteration $x^{gb}(n)$, the optimal values of weighting matrices **Q**, R and controller gain **K** are obtained.

4 Results and observations

In the previous section, the derivation of the LQR-PSO and LQRI-PSO controller weighting matrices [**Q**], [R], and controller gain [**K**] using the particle swarm optimization (PSO) technique are presented. This section focuses on evaluating the performance of the proposed LQR-PSO and LQRI-PSO controllers. The absolute maximum peak ($|M_p|$) and error indices for the cart position (x_c) relative to the reference input are computed and summarized in respective tables. All $|M_p|$ values are derived with the reference set to zero. The effectiveness of these controllers is analyzed through simulations of a single inverted pendulum (SIP), followed by experimental validation and tests related to robustness of the proposed controllers with unmodeled dynamics (Fig. 5).

4.1 Simulation study

The analysis is carried out for simulation of single inverted pendulum (SIP), and the experimental validations are conducted and the observation are presented. For the proposed LQR-PSO controller, the optimal values of [**Q**] = diag (154.5367, 395.1028, 0.4908, 0.2787), [R] = 0.037, and [**K**] = [−64.6554, 158.7123, −52.1483, 17.6735]. For the LQRI-PSO controller, the optimal values of [**Q**] = diag(140, 440, 0.1691, 0.4999, 0.0100), [R] = 0.036, and [**K**] = [−62.2336, 164.4446, −53.0049, 18.1426, −0.5227]. For the optimal values of [**Q**], [R], and controller gain [**K**], the simulations using both the controller gains of the LQR-PSO and LQRI-PSO controllers are conducted. To balance the pendulum in the inverted position, the cart is provided with a square waveform with an amplitude of ±20 mm, with a time period of 20 s. For the PSO algorithm, the number of particles N is considered as 30, and the maximum number of iterations $t_{\max}$ is 50, c_1 and c_1 are 1.494, and w is 0.729 [51]. The responses for the experimental setup are presented in Figs. 6 and 7. The error indices like integral square error (ISE), integral absolute error (IAE), integral time square error (ITSE), and integral time absolute error (ITAE) are presented in Table 2. Table 2 shows that both the proposed controllers show lower values of ISE, IAE, ITSE, and ITAE for simulation study. Although Table 2 presents only the mean values of the error indices for the tracking simulations under square-type reference input, the standard deviation was also computed across multiple trials to assess consistency. The results showed that all standard

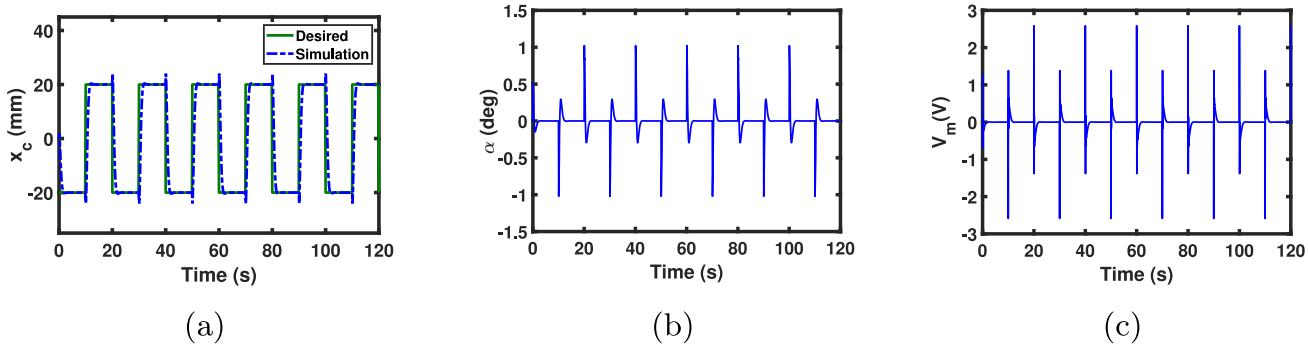


Fig. 6 Simulation responses achieved through the proposed LQR-PSO controller, for the square reference input; **a** variations in the cart position x_c; **b** variations in pendulum angle α; **c** variations in the input voltage V_m

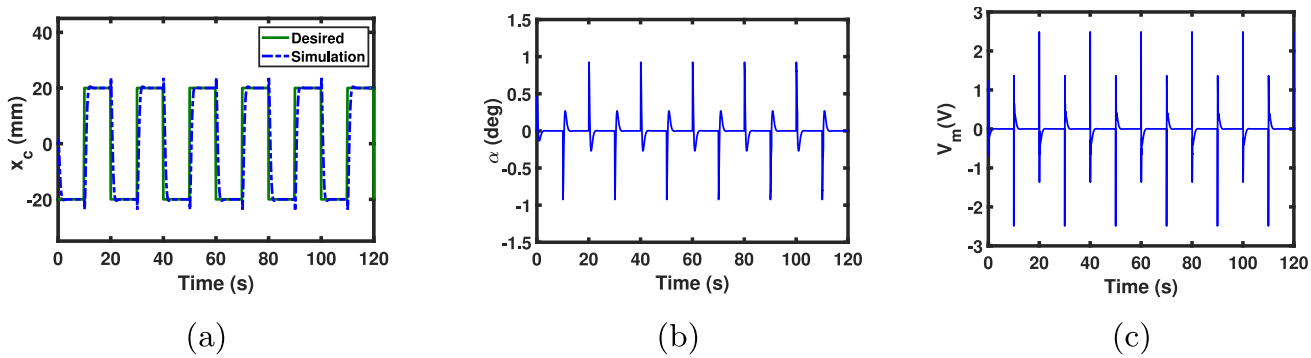


Fig. 7 Simulation response achieved through the proposed LQRI-PSO controller, for the square reference input; **a** variations in the cart position x_c, with respect to the desired square reference input; **b** variations in pendulum angle α; **c** variations in the input voltage V_m

deviations were relatively low across both LQR-PSO and LQRI-PSO controllers. For instance, the ISE and IAE metrics exhibited standard deviations below 0.04, while ITSE and ITAE remained within a tight range of approximately 1.7–3.0. These low variations indicate that the controller performance was stable and repeatable across different simulation runs, thereby reinforcing the reliability of the presented mean values.

However, it is observed that the LQRI-PSO controller shows superior performance as compared to the LQR-PSO controller, which is evident from the lower error indices values. Now to further check the effectiveness of the proposed controllers, the simulations are conducted using the sinusoid- and sawtooth-type reference inputs. The optimal weighting matrices [**Q**], [R], and controller gains [**K**] are obtained using PSO algorithm for both the sinusoid- and sawtooth-type reference inputs, the optimal controller gains for square-, sinusoid-, and sawtooth-type reference inputs are presented in Table 4. The convergence curves achieved while obtaining optimal controller gains for various reference inputs are presented in Fig. 8. It is also observed that for both the proposed controllers and all desired signals, the convergence occurs $\leq$13 iterations. Futhermore, the absolute maximum peak values $|M_p|$ values for V_m, x_c, and α while using various reference inputs are presented in Table 3. Table 3 shows that both the LQR-PSO and LQRI-PSO controller perform effectively with lower values of $|M_p|$. However, the LQRI-PSO controller gives better performance as compared to LQR-PSO controller.

To further test the efficacy of the proposed LQR-PSO and LQRI-PSO controllers, the reference inputs like sinusoidal and sawtooth inputs are given to the system, and for these reference inputs. By using the method presented in Sect. 3.3, the respective controller gains [**K**] are obtained for both the LQR-PSO and LQRI-PSO controllers for respective reference inputs. The optimal controller gains for square-, sinusoid-, and sawtooth-type reference inputs for LQR-PSO and LQRI-PSO controllers are presented in Table 4. The convergence curves for obtaining the optimal controller gains are presented in Fig. 8. Using the controller gains presented in Table 4, experimental validations are conducted on the experimental setup presented in Fig. 13a. To analyze the influence of PSO hyperparameters on control performance, a parameter sweep was conducted over the inertia weight (w) and acceleration coefficients (c_1, c_2). Table 5 reports the corresponding Mean Squared Error (MSE) in the cart position x, expressed in mm^2. The configuration $w = 0.7$, $c_1 = 1.5$, and $c_2 = 1.5$ resulted in the lowest MSE of 0.43, indicat-

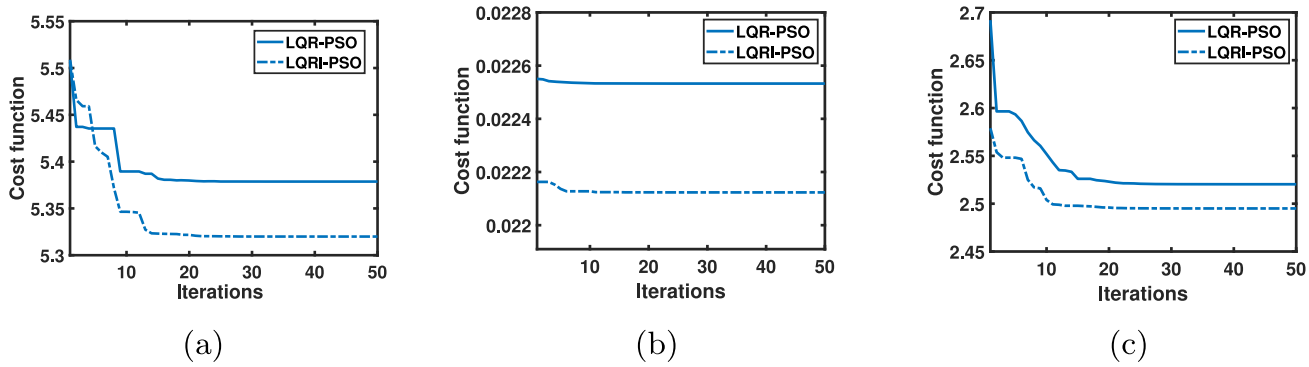


Fig. 8 Convergence curves for the LQR-PSO and LQRI-PSO controllers for various reference inputs: **a** square; **b** sinusoidal; **c** sawtooth

Table 3 Absolute maximum peak $|M_p|$ obtained through simulations of linear inverted pendulum system from various reference inputs from $t = 0$ s to $t = 120$ s

Reference input	LQR-PSO			LQRI-PSO		
	V_m (V)	x_c (mm)	α (deg)	V_m (V)	x_c (mm)	α (deg)
Square	2.5862	24.0090	1.0215	2.4892	23.6380	0.9248
Sinusoidal	0.047742	19.9557	0.042880	0.047722	19.9440	0.042680
Sawtooth	2.5097	22.3987	0.9422	2.4908	22.3576	0.9298

Table 4 Optimal controller gains for square-, sinusoid-, and sawtooth-type reference input for LQR-PSO and LQRI-PSO controller

Reference input	LQR-PSO				LQRI-PSO				
	K_1	K_2	K_3	K_4	K_1	K_2	K_3	K_4	K_5
Square	−64.6554	158.7123	−52.1483	17.6735	−62.2336	164.4446	−53.0049	18.1426	−0.5227
Sinusoidal	−70.7107	157.2181	−54.7952	17.3974	−68.4762	151.9806	−53.1123	16.9215	−5.1988
Sawtooth	−62.3610	163.2345	−53.0059	18.1659	−61.8931	163.0442	−52.7876	17.8252	−0.5199

Table 5 Effect of PSO Hyperparameters on MSE of cart position x (in mm)

Inertia weight (w)	c_1	c_2	MSE (mm^2)
0.4	1.0	1.0	0.71
0.7	1.5	1.5	0.43
0.9	2.0	2.0	0.58
1.2	2.5	2.5	0.85

ing optimal convergence and improved estimation accuracy. These results highlight the critical role of PSO hyperparameters in fine-tuning controller performance for precise motion regulation.

4.2 Comparative analysis and applications in inverted pendulums, cubli rovers, and monowheel robot

The proposed linear quadratic regulator–particle swarm optimization (LQR-PSO) and linear quadratic regulator with integral–particle swarm optimization (LQRI-PSO) controllers are tested on various inverted pendulum models [8]-[11], including Cubli rovers [46] and monowheel robots [47]. The Cubli rovers and the monowheel robots platform are used for the comparative analysis, as these models work on the principles of inverted pendulum [46, 47].

The proposed controllers are tested on various other inverted pendulum models [8–11]. These models include the classic cart–pendulum system, which has been linearized and presented as a state-space model for control design. In [8], a Newton–Raphson method-based LQR controller is proposed, which is compared with exponential LQR controller and genetic algorithm-based LQR controller. The effectiveness of the proposed LQR-PSO and LQRI-PSO controllers is tested by providing the desired reference input and the cart tracks the reference input effectively, and the results obtained for the inverted pendulum model 1 in [8] are presented in Fig. 9. Similarly, the results obtained for the inverted pendulum models 2–4 are presented in Fig. 9. It is obtained from the results that both the LQR-PSO and LQRI-PSO controllers effectively track the reference inputs.

Cubli rovers are reaction wheel-based inverted pendulum systems [46] (Fig. 10a), and these are utilized to test the proposed controllers. The results obtained for the LQR-PSO and LQRI-PSO controllers are presented in Fig. 11 while

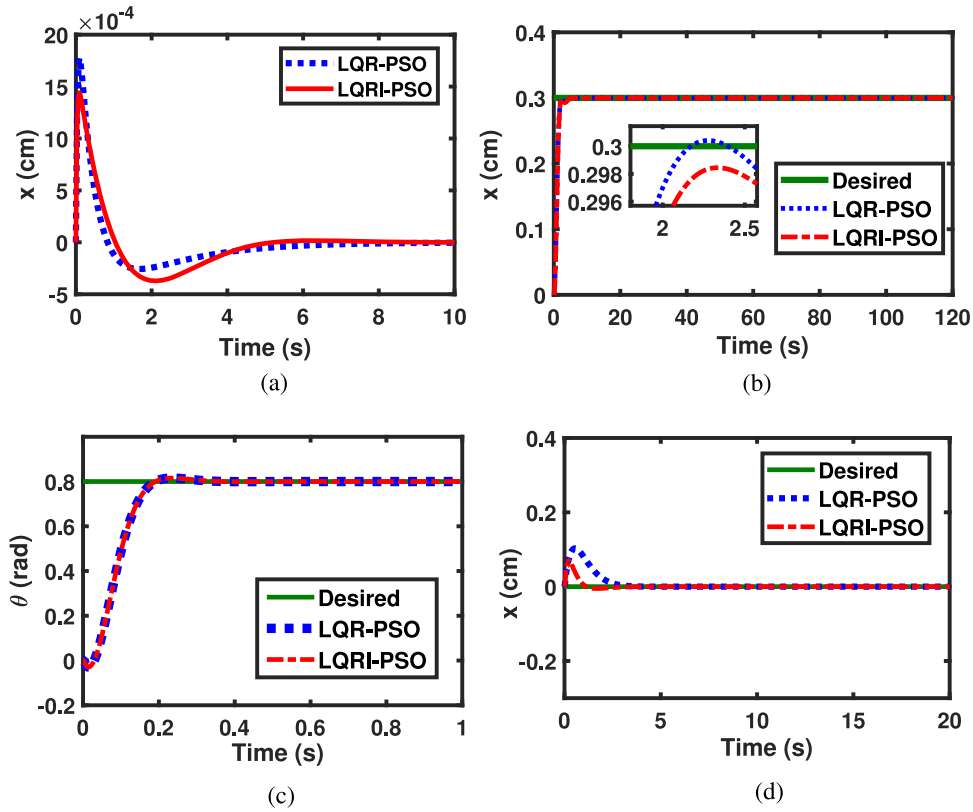


Fig. 9 Performance of the proposed LQR-PSO and LQRI-PSO controllers on inverted pendulum models: **a** Model 1 [8]; **b** Model 2 [9]; **c** Model 3 [10]; Model 4 [11]

Table 6 Performance indices for the comparative analysis of the proposed LQR-PSO and LQRI-PSO controllers in inverted pendulum models 1 and 2

Inverted pendulum models	Controllers	Settling time (t_s(s))	Absolute maximum peak ($\|M_p\|$(cm))
Model 1 [8]	E-LQR [8]	> 6 (approx.)	0.098
	GA-LQR [8]	3 (approx.)	0.039
	NR-LQR [8]	> 6 (approx.)	0.056
	LQR-PSO	2.824	1.771×10^{-3}
	LQRI-PSO	2.436	1.439×10^{-3}
Model 2 [9]	PID [9]	4	–
	FV-PID [9]	2	–
	LQR-PSO	1.75	–
	LQRI-PSO	1.80	–

Table 7 Error indices for the comparative analysis of the proposed LQR-PSO and LQRI-PSO controllers in inverted pendulum models 3 and 4

Inverted pendulum models	Controllers	ISE	IAE	ITSE	ITAE
Model 3 [10]	PID [10]	0.610×10^{-2}	1.8×10^{-2}	6.786×10^{-5}	9.0×10^{-4}
	IAC [10]	2.471×10^{-6}	1.5×10^{-3}	1.223×10^{-6}	7.0×10^{-4}
	NFC [10]	1.368×10^{-5}	3.6×10^{-3}	6.781×10^{-6}	8.0×10^{-3}
	LQR-PSO	3.832×10^{-6}	5.2×10^{-4}	8.548×10^{-7}	9.4×10^{-5}
	LQRI-PSO	1.415×10^{-6}	1.3×10^{-4}	5.716×10^{-7}	6.2×10^{-5}
Model 4 [11]	PID [11]	–	–	0.609	3.88
	VSC [11]	–	–	0.225	2.85
	LQR-PSO	3.057×10^{-6}	2.5×10^{-3}	2.382×10^{-6}	0.0024
	LQRI-PSO	5.648×10^{-7}	7.7×10^{-4}	1.985×10^{-7}	0.0005

Table 8 Performance indices for the comparative analysis of the proposed LQR-PSO and LQRI-PSO controllers in Cubli rover and monowheel system

System	Parameters	Controllers	Settling time (t_s(s))	Absolute maximum peak ($\|M_p\|$)
Cubli rover [46]	$\underbrace{w_x, w_y, w_z}_{\text{rad/s}}$	APPC [46]	$\underbrace{\geq 1.8, \geq 1.85, \geq 1.9}_{\text{approx.}}$	$\underbrace{0.98, 1.25, 1.05}_{\text{approx.}}$
		LQR-PSO	0.132, 1.062, 1.028	0.0063, 0.00042, 0.001137
		LQRI-PSO	0.136, 1.058, 1.017	0.0068, 0.00031, 0.001131
Monowheel system [47]	x (m)	LQR [47]	2.5 (approx.)	1.025 (approx.)
		LQR-PSO (step)	3.08	1.007
		LQRI-PSO (step)	3.14	1.003

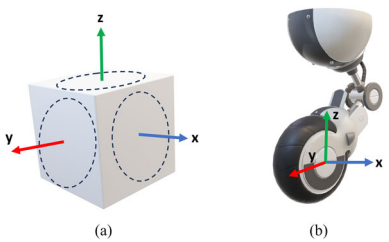


Fig. 10 Applications of inverted pendulum-inspired systems: **a** Cubli rovers [46]; **b** monowheel robots [47]

maintainig the angular velocities w_x, w_y, and w_z to 0 in x, y,and z direction, respectively, so that the Cubli rover balances in upright position [46]. It is observed that both the LQR-PSO and LQR-PSO controllers effectively track the desired reference.

Monowheel robots [47], another type of inverted pendulum model (Fig. 10b), have also been tested with the LQR-PSO and LQRI-PSO controllers. These robots present unique challenges due to their single-wheel design, which requires better control to maintain balance and stability. For

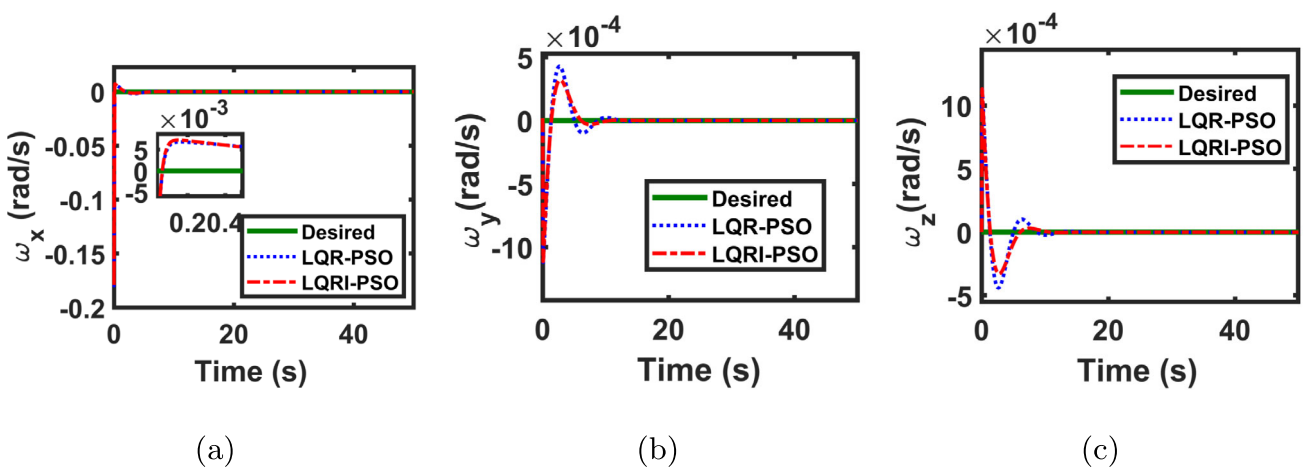
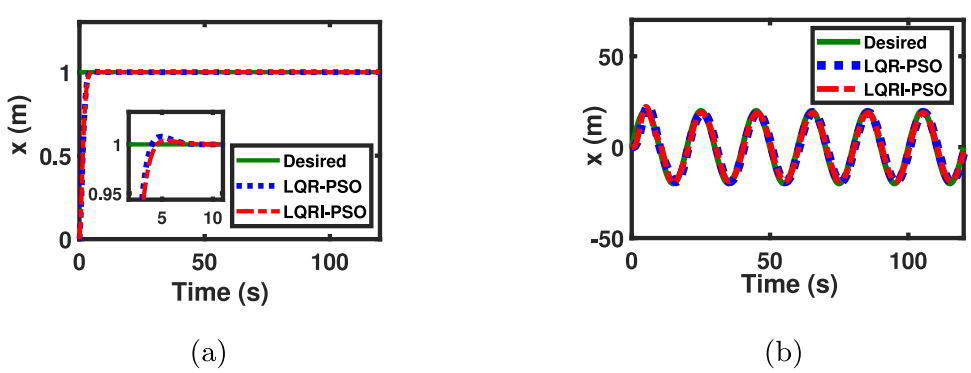


Fig. 11 Performance of the proposed LQR-PSO and LQRI-PSO controllers to Cubli rovers in asteroid [46] while maintaining the angular velocities to 0 for: **a** w_x; **b** w_y; and **c** w_z

Fig. 12 Performance of the proposed LQR-PSO and LQRI-PSO controllers to monowheel robot [47] while tracking the desired reference inputs: **a** unit step; **b** sinusoidal



the monowheel, the results obtained are presented in Fig. 12. It is observed that the proposed controllers effectively track the desired reference inputs. The comparative results for the Cubli rover and monowheel robots are presented in Table 8.

The performance indices like ISE, IAE, ITSE, and ITAE for the proposed controllers with applications in inverted pendulum models in [8–11], Cubli rovers in [46], and monowheel applications in [47] are presented in Tables 7, 6, and 8. The proposed controllers are tested with different desired reference inputs and the results corroborate to the fact the proposed LQR-PSO and LQRI-PSO controllers effectively track the desired inputs. Thus, these comparative analysis testify the fact that the proposed LQR-PSO and LQRI-PSO controllers can be used in other inverted pendulum models and in inverted pendulum applications.

4.3 Experimental study

To test the efficacy of the proposed LQR-PSO controller, the optimal controller gains are tested on a experimental single-inverted-pendulum setup as presented in Fig. 13a.

The Quanser Linear Servo Base Unit with Inverted Pendulum (IP02) consists of a linear cart driven by a DC motor and an attached inverted pendulum, forming a nonlinear underactuated system. The setup is controlled and simulated using MATLAB/Simulink with the QUARC add-on for real-time implementation. Data acquisition is handled through interfaces such as Q2-USB with a sampling time of 1 millisecond, Q8-USB, or QPID/QPIDe, and the system is powered by the VoltPAQ-X1 amplifier. It includes a dynamic model and pre-designed Simulink controllers.

The hardware setup consists of a pendulum, a linear moving cart, a power amplifier, data acquisition system, and a PC. For square-type reference input with an amplitude of ± 20 mm, with a time period of 20 s, experimental tests are conducted using the controller gains for square-type reference input as presented in Table 4. The experimental data reveals an initial spike in the outputs of V_m, x_c, and α. This spike occurs as a result of manually rotating the pendulum from its gantry (downward inverted) position to the inverted position, as illustrated in Fig. 13a. Thus, the error indices presented in Table 9, which are calculated from $t = 30$ s to $t = 120$ s. All the experimental analyses are conducted using square, sinusoidal, and sawtooth reference inputs, with comparative data presented in Table 9. For clarity, selected results are shown in the figures.

The error indices like integral squared error (ISE), integral absolute error (IAE), integral time squared error (ITSE), and integral time absolute error (ITAE) for square-type tracking input for cart position x_c is presented in Table 9. Table 9 shows that the error indices for the LQRI-PSO controller are lower as compared to the LQR-PSO controller. It is observed that the proposed LQR-PSO and the LQRI-PSO controller effectively track the given desired square input. As presented in Table 9, for the LQR-PSO controller, the $|M_p|$ for the V_m is 7.2957 V, similarly the $|M_p|$ for x_c is 28.6404 mm, and $|M_p|$ for α is 1.3031^o. Similarly, for the LQRI-PSO controller, the $|M_p|$ for the V_m is 7.8180 V, similarly the $|M_p|$ for x_c is 26.5021 mm, and $|M_p|$ for α is 1.2305^o. Table 9 and the results in Fig. 14 show that both the LQR-PSO and LQRI-PSO controller are effective in controlling the inverted pendulum. It must be noted that the pendulum remains upright during the experiment/simulation as the α stays very close to zero. However, the LQRI-PSO controller surpasses the LQR-PSO controller, as the error indices for the LQRI-PSO controller are lower as compared to the LQR-PSO controller. Also Table 9 shows that, for the square-type reference tracking, the maximum peak $|M_p|$ for the cart position x_c and pendulum angle α are lower for the LQRI-PSO controller.

4.4 Robustness analysis for various reference inputs

To further evaluate the robustness of the LQR-PSO and LQRI-PSO controllers, the analysis is extended to include sinusoid- and sawtooth-type reference inputs. As outlined in Sect. 3.3, the PSO method is employed to derive the weighting matrices $\mathbf{Q}$ and R, as well as the controller gains $\mathbf{K}$. These gains are then utilized for the LQR-PSO and LQRI-PSO controllers in experimental trials, with the findings subsequently presented. For the sinusoid-type reference input, the optimal gains are presented in Table 4. In Table 9, the performance metrics for the two controllers are provided. For the LQR-PSO controller, the $|M_p|$ for the variables V_m, x_c, and α are 3.1331 V, 27.6395 mm, and 0.7031^o, respectively. Similarly, for the LQRI-PSO controller, the corresponding values are 5.0770 V, 25.0234 mm, and 0.8789^o for the variables V_m, x_c, and α. These metrics provide insights into the performance of each controller in terms of absolute maximum peak, indicating how much each variable exceeds its steady-state value during the control process. The comparison between the LQR-PSO and LQRI-PSO controllers is further examined through the analysis of their performance in tracking a sinusoidal reference input, as depicted in Fig. 14. Both controllers demonstrate effective tracking of the sinusoidal reference input, indicating their capability to regulate the system dynamics according to the desired input. To delve deeper into their performance, various error indices such as ISE, IAE, ITSE, and ITAE are evaluated and presented in Table 9 for the sinusoidal reference input. The results presented in Table 9 apparently show that the LQRI-PSO controller generally outperforms the LQR-PSO controller in terms of ISE and ITSE values, indicating lower ISE and ITSE over time. On the other hand, while the IAE and ITAE values for both controllers are closer to each other, implying similar abso-

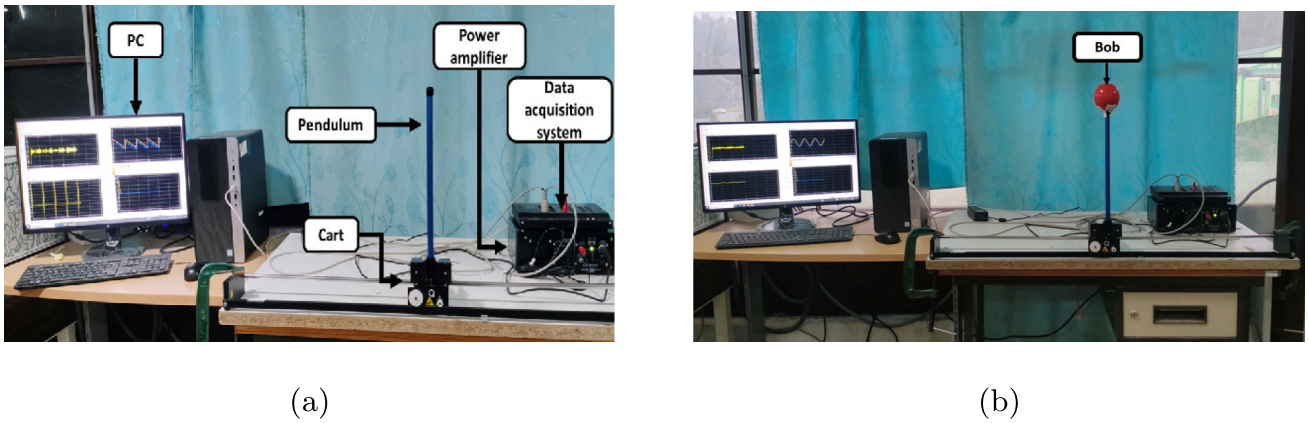


Fig. 13 Experimental setup for the linear single inverted pendulum on a moving cart setup: **a** nominal system; **b** system with bob (external weight)

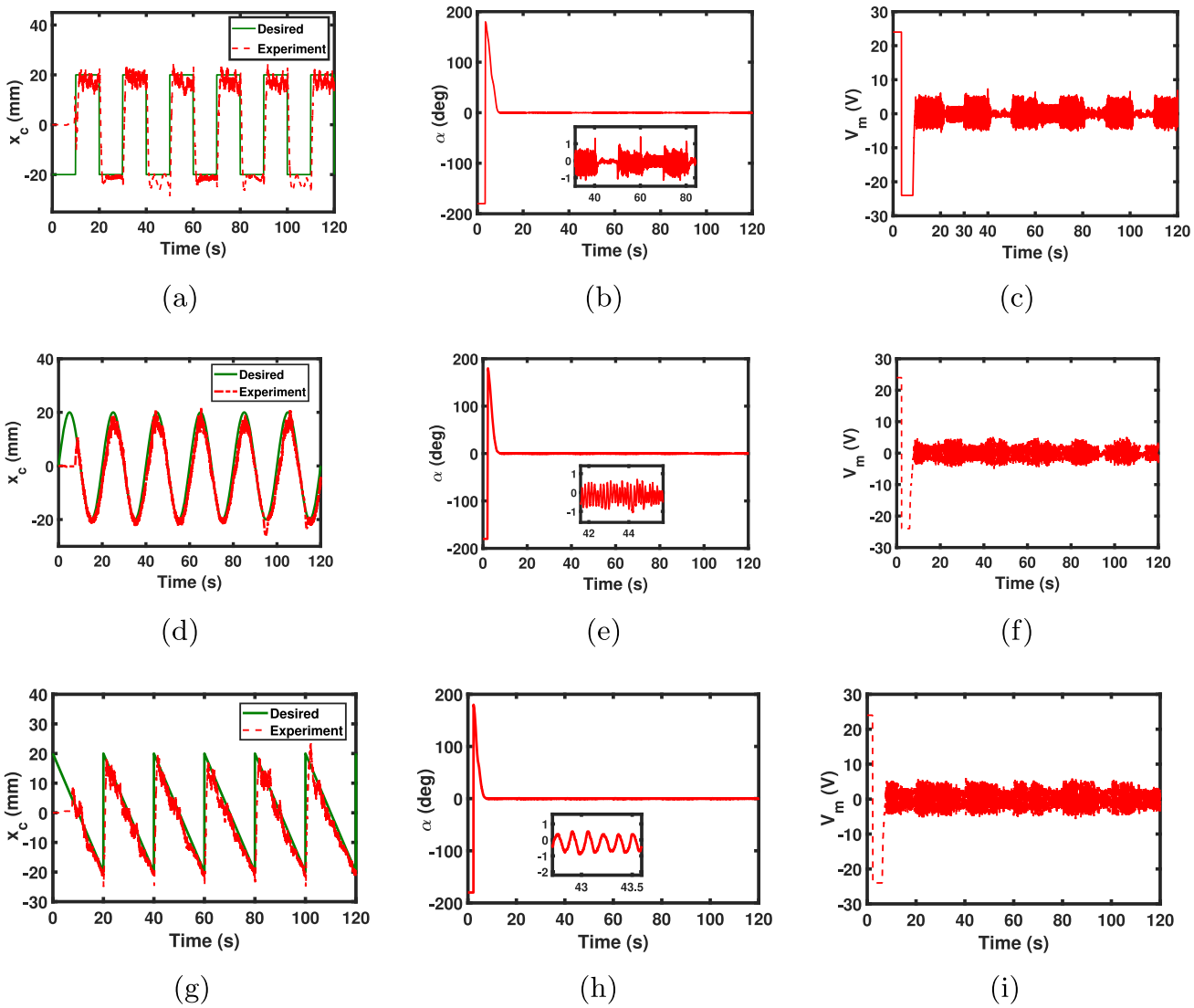


Fig. 14 Experimental responses achieved through the proposed controllers for the nominal system: the LQR-PSO controller for the square reference input (**a**–**c**); LQRI-PSO controller for the sinusoidal reference input (**d**–**f**); LQR-PSO controller for the sawtooth reference input (**g**–**i**); variations in (**a**, **d**, **g**) the cart position x_c; **b**, **e**, **h** the pendulum angle α; **c**, **f**, **i** the input voltage V_m

Table 9 Experimental results on error indices (ISE, IAE, ITSE, ITAE) and the maximum peak $|M_p|$ values of the cart position (x_c), the pendulum angle (α), and the input voltage (V_m) for different reference inputs ($t = 30$ s to $t = 120$ s)

System	Performance indices	Square		Sinusoidal		Sawtooth	
		LQR-PSO	LQRI-PSO	LQR-PSO	LQRI-PSO	LQR-PSO	LQRI-PSO
Nominal system	ISE	2.747000	2.680800	0.455000	0.434600	1.346600	1.146100
	IAE	9.289500	8.018200	5.106400	5.134700	5.415600	4.177500
	ITSE	194.4486	191.6998	35.03920	33.27330	95.94510	81.25570
	ITAE	678.4596	595.0552	391.3119	391.9642	400.4582	304.4550
	x_c (mm)	28.64040	26.50210	27.63950	25.02340	24.81870	23.93150
	α (deg)	1.303100	1.230500	0.703100	0.878900	1.230500	1.406200
	V_m (V)	7.295700	7.818000	3.133100	5.077000	5.984200	5.907000
System with bob	ISE	2.379500	2.350100	0.661600	0.645200	1.302700	1.192500
	IAE	6.534300	6.220600	6.874000	6.748700	6.319300	5.564900
	ITSE	167.8197	165.1483	50.74350	49.15110	92.02710	84.90540
	ITAE	467.4823	455.9359	523.2104	510.7245	466.9145	414.7726
	x_c (mm)	28.16270	26.18360	28.29920	27.63950	27.36650	26.38830
	α (deg)	1.669900	1.394300	0.713400	0.674100	1.142600	1.054700
	V_m (V)	5.472000	5.491700	3.800700	3.133100	5.005800	4.655200

Table 10 Experimental results on error indices (ISE, IAE, ITSE, ITAE) and the maximum peak $|M_p|$ values of the cart position (x_c), the pendulum angle (α), and the input voltage (V_m) for different reference inputs ($t = 30$ s to $t = 120$ s)

System	Performance indices	Square		Sinusoidal		Sawtooth	
		LQR-PSO	LQRI-PSO	LQR-PSO	LQRI-PSO	LQR-PSO	LQRI-PSO
Nominal system	ISE	2.761800	2.689900	0.459300	0.437200	1.351700	1.153800
	IAE	9.306000	8.036500	5.121700	5.142100	5.429900	4.189600
	ITSE	194.8931	191.3510	35.12930	33.29500	96.03340	81.37290
	ITAE	681.0534	597.6449	391.8725	392.1153	401.2014	305.2748
	x_c (mm)	28.67290	26.52300	27.65180	25.04100	24.83140	23.94890
	α (deg)	1.308400	1.236900	0.707000	0.883200	1.236400	1.411300
	V_m (V)	7.313800	7.831100	3.140200	5.083900	5.996400	5.913500

lute errors, the LQRI-PSO controller still maintains a slight advantage.

Similarly, the comparative analysis between the LQR-PSO and LQRI-PSO controllers extends to their performance in tracking a sawtooth-type reference input, as illustrated in Fig. 14. For the sawtooth-type reference input, the optimal gains are presented in Table 4. Figure 14 shows that both controllers exhibit adept tracking capabilities with respect to the sawtooth-type reference input. To delve deeper into their performance, error indices such as ISE, IAE, ITSE, and ITAE are computed and presented in Table 9 specifically for the sawtooth-type reference input. Analysis of these results reveals that the LQRI-PSO controller generally showcases superior performance over the LQR-PSO controller, particularly in terms of lower ISE, IAE, ITSE, and ITAE values.

The performance evaluation of the proposed LQR-PSO and LQRI-PSO controllers for the single-inverted-pendulum system across diverse reference inputs including square, sinusoidal, and sawtooth inputs reveals notable insights into their effectiveness. Through comprehensive analysis and comparison, it becomes evident that both controllers exhibit commendable tracking capabilities across all reference input types. They demonstrate the ability to regulate the system dynamics in response to each input's varying characteristics, maintaining stability and minimizing error accumulation over time. Each controller showcases strengths across different performance metrics, such as absolute maximum peak ($|M_p|$), ISE, IAE, ITSE, and ITAE values. The LQRI-PSO controller consistently emerges as a slightly superior option, particularly in terms of reducing integral errors and enhancing overall tracking capability. This comparative assessment underscores the robustness and versatility of both the LQR-PSO and LQRI-PSO controllers in handling diverse reference input inputs, reaffirming their potential for effective real-world applications in controlling single-inverted-pendulum systems and their applications across various operating con-

ditions. To evaluate the performance and robustness of the proposed controllers, the same optimized gains were used for all types of reference inputs, including square, sinusoidal, and sawtooth. For the LQR-PSO controller, the optimal gain vector was defined as $\mathbf{K}$ = [−64.6554 158.7123 −52.1483 17.6735]. Similarly, for the LQRI-PSO controller, the gain vector used was $\mathbf{K}$ = [−62.2336 164.4446 −53.0049 18.1426 −0.5227], where the fifth term corresponds to the integral action. These fixed gains were applied across different operating conditions to examine the controller's ability to maintain performance and ensure robustness against variations in reference inputs and model uncertainties. The error metrics obtained are presented in Table 10.

4.5 Analysis with unmodeled dynamics

To thoroughly test the robustness of the LQR-PSO and LQRI-PSO controllers, the experimental setup is subjected to square-, sinusoid-, and sawtooth-type input inputs while introducing unmodeled dynamics externally (bob) to the inverted pendulum, a brief overview is presented in Fig. 15. These dynamics simulate real-world disturbances and uncertainties that were not accounted for in the original model, thus providing a challenge for the controllers, using the same setup described earlier, as shown in Fig. 13a, and the optimal controller gains, as listed in Table 4. The proposed LQR-PSO and LQRI-PSO controllers are tested on the single-inverted-pendulum setup with external disturbance (bob) having weight 19.48 gs, which is attached on top of the pendulum, as presented in Fig. 13b. This bob acts as an unmodeled addition to the existing system. Now, the proposed controllers are tested using the weighing matrices $\mathbf{Q}$ and R, and the controller gains $\mathbf{K}$ obtained in Sect. 4 for the respective square-, sinusoid-, and sawtooth-type reference input. For the square-type reference input, for the proposed LQR-PSO controller, the optimal values of [$\mathbf{Q}$] = diag(154.5367, 395.1028, 0.4908, 0.2787), [R] = 0.037, and [$\mathbf{K}$] = [−64.6554, 158.7123, −52.1483, 17.6735]. For the LQRI-PSO controller, the optimal values of [$\mathbf{Q}$] = diag(140, 440, 0.1691, 0.4999, 0.0100), [R] = 0.036, and [$\mathbf{K}$] = [−62.2336, 164.4446, −53.0049, 18.1426, −0.5227]. These controller gain values are used on the experimental setup with bob as presented in Fig. 13b, and the results are obtained as presented in Fig. 16. It is observed from the results that, though we have a modified system, both the LQR-PSO and the LQRI-PSO controllers effectively track the reference input, and also the error indices for the square-type reference input for both the controllers are tabulated in Table 9. Also the absolute maximum peak $|M_p|$ for V_m, x_c, and α for square-type reference input are presented in Table 9. Table 9 shows that both the LQR-PSO and LQRI-PSO controllers have moderate $|M_p|$ values despite the bob. However, the LQRI-PSO controller outperforms the LQR-PSO controller, as the $|M_p|$ values for x_c and α are lower for the LQRI-PSO controller for square-type reference input.

For the sinusoidal reference input, the proposed LQR-PSO controller utilizes optimal values of [$\mathbf{Q}$] = diag(169.9,350, 0.0360, 0.0029), [R] = 0.034, and [$\mathbf{K}$] = [−70.7107, 157.2181, −54.7952, 17.3974]. Similarly, for the LQRI-PSO controller, the optimal values of [$\mathbf{Q}$] = diag(155.98, 349.12, 0.1975, 0.042, 0.99), [R] = 0.037, and [$\mathbf{K}$] = [−68.4762, 151.9806, −53.1123, 16.9215, −5.1988]. These controller parameters are applied to the experimental setup featuring a bob, as depicted in Fig. 13b, which yields results shown in Fig. 16. Analysis of the results reveals that despite modifications to the system, both the LQR-PSO and LQRI-PSO controllers effectively track the sinusoidal reference input. Error indices for the sinusoidal reference input for both controllers are tabulated in Table 9. Additionally, Table 9 presents the absolute maximum peaks $|M_p|$ for V_m, x_c, and α. Notably, both controllers exhibit moderate $|M_p|$ values despite the presence of the bob. However, the LQRI-PSO controller outperforms the LQR-PSO controller, as evidenced by lower $|M_p|$ values with the sinusoidal reference input, as indicated in Table 9.

For the sawtooth-type reference input, the proposed LQR-PSO controller is optimized with values such as [$\mathbf{Q}$] = diag(140,418.11, 0.4990, 0.5), [R] = 0.0360, and [$\mathbf{K}$] = [−62.3610, 163.2345, −53.0059, 18.1659]. Similarly, the LQRI-PSO controller adopts optimal values like [$\mathbf{Q}$] = diag(140, 440, 0.5, 0.3517, 0.01), [R] = 0.037, and [$\mathbf{K}$] = [−61.8931, 163.0442, −52.7876, 17.8252, −0.5199]. These parameter sets are implemented in the experimental setup featuring a bob, as shown in Fig. 13b, producing results depicted in Fig. 16. Evaluation of the results indicates that despite system modifications, both controllers adeptly follow the sawtooth-type reference input. Error indices for this input type are tabulated in Table 9 for both controllers. Furthermore, Table 9 provides absolute maximum peaks $|M_p|$ for V_m, x_c, and α. Remarkably, despite the bob's presence, both controllers demonstrate moderate $|M_p|$ values. However, the LQRI-PSO controller exhibits superior performance compared to the LQR-PSO controller, evidenced by lower $|M_p|$ values with the sawtooth-type reference input, as highlighted in Table 9.

In conclusion, the investigation into the performance of the proposed LQR-PSO and LQRI-PSO controllers on a single-inverted-pendulum system, modified with a bob, reveals promising results across different reference input waveforms. For the square-type reference input, both controllers effectively track the desired trajectory, as evidenced by the analysis of error indices and absolute maximum peaks. Despite the system modifications, moderate $|M_p|$ values are observed, indicating reasonable stability. Transitioning to the sinusoidal reference input, similar outcomes are observed, with both controllers demonstrating adept

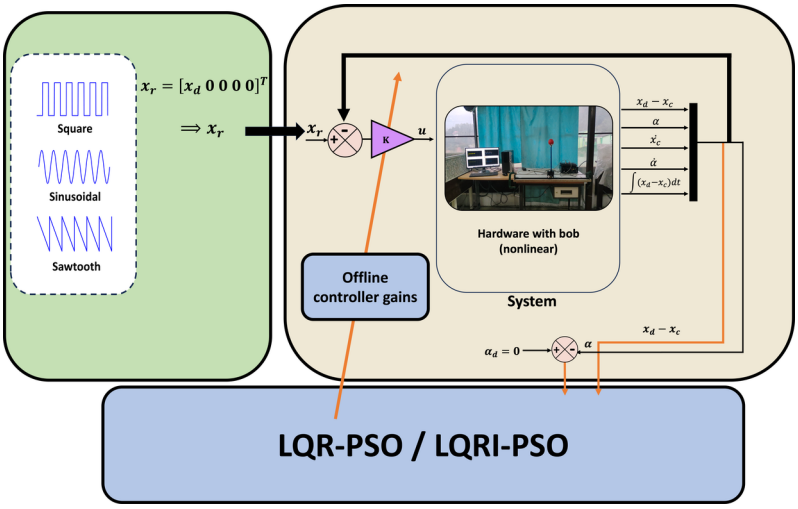


Fig. 15 An outline of the LQRI-PSO controller design for the system with external influence (bob) using different reference inputs

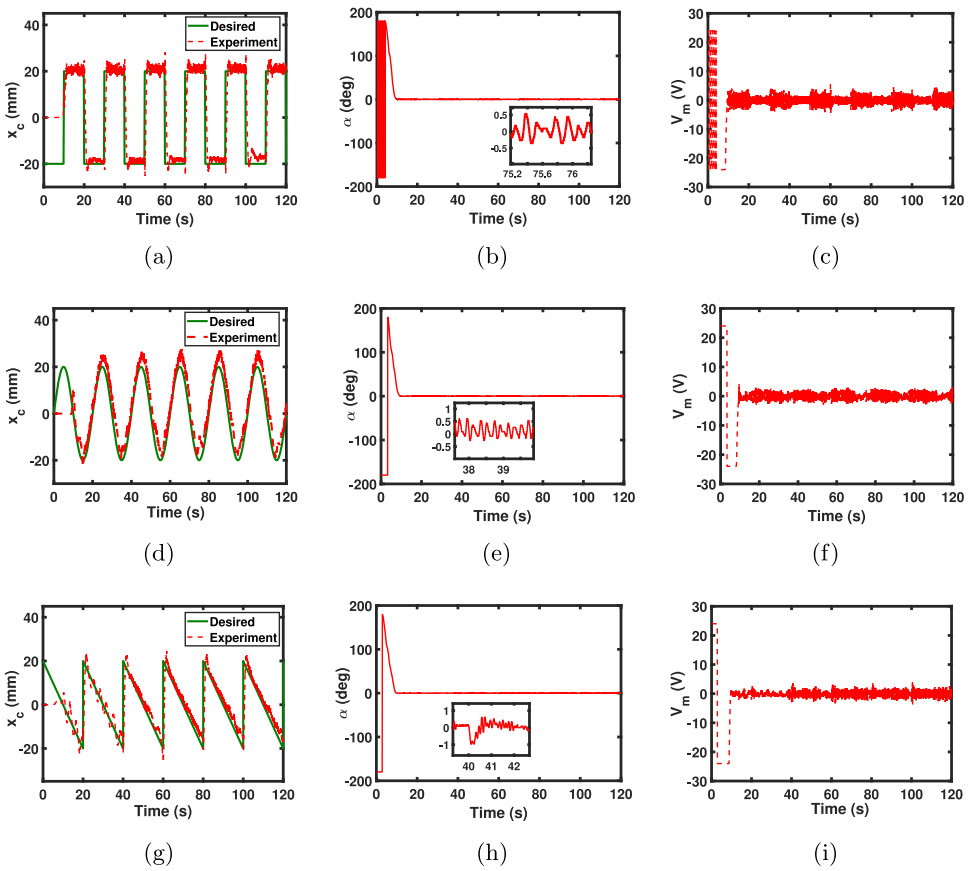


Fig. 16 Experimental responses achieved through the proposed controllers for the system with bob (external weight): the LQRI-PSO controller for the square reference input (**a**–**c**); LQR-PSO controller for the sinusoidal reference input (**d**–**f**); LQRI-PSO controller for the sawtooth reference input (**g**–**i**); variations in (**a**, **d**, **g**) the cart position x_c; **b**, **e**, **h** the pendulum angle α; **c**, **f**, **i** the input voltage V_m

tracking capabilities and maintaining moderate peak values. Furthermore, when tested with a sawtooth-type reference input, the controllers exhibit consistent performance, ensuring precise tracking of the desired motion. Notably, the LQRI-PSO controller consistently outperforms the LQR-PSO controller across all waveform scenarios, showcasing its enhanced ability to minimize peak values and maintain stability. These findings underscore the effectiveness of the LQRI-PSO approach in optimizing controller parameters for improved performance in complex dynamical systems like the single inverted pendulum with a bob.

5 Conclusion

In conclusion, this work provides a thorough validation of the proposed linear quadratic regulator with particle swarm optimization (LQR-PSO) and linear quadratic regulator with

integrator based on PSO (LQRI-PSO) controllers for managing the dynamics of the inverted pendulum-inspired systems. By employing PSO, optimal controller gains were determined, ensuring enhanced performances. The controllers were rigorously tested against various reference inputs and applied to diverse inverted pendulum models, including Cubli rovers and monowheel robots, demonstrating their robustness and adaptability. Comparative studies with other advanced controllers such as neuro-fuzzy controller, indirect adaptive controller, genetic algorithm-based LQR controller, Newton–Raphson method-based LQR controller, exponential LQR controller, adaptive prescribed performance controller (APPC), and proportional–derivative–integral (PID) controller further highlight the applicability of the proposed solutions. Extensive simulations and experimental analysis, including scenarios with added external weights and unmodeled dynamics, underscore the controllers' robustness in maintaining system stability and minimizing errors. Performance indices such as the ISE, IAE, ITSE, ITAE, and the absolute maximum peak reinforce the efficacy of the proposed controllers for various applications involving inverted pendulums. The proposed controllers are based on the linearized model of the inverted pendulum-inspired systems. A future scope includes design of a controller while keeping the nonlinear system model.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Ethics statement The authors declare that the research was conducted ethically and in accordance with all relevant guidelines. No human or animal subjects were involved in this study.

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